

Supplementary Online Content

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eTable 1. Detailed Baseline Characteristics of Participants, According to Randomized Vitamin D3 and Placebo Groups

eTable 2. Mean Difference in Change Since Baseline in PHQ-8 Score Comparing Vitamin D3 and Placebo Groups, According to Baseline Subgroups

Description of Results. From Sub-Group Analyses in Figure 3 and eTable 2

eTable 3. Adjusted Means at Baseline and Mean Change (95% CI) in PHQ-8 Scores at Each Year Since Randomization Compared to Baseline, according to Vitamin D3 and Placebo Groups, Censoring PHQ-8 Scores after Initiation of Antidepressants

eTable 4. Participant-Reported Adherence with the Vitamin D3 and Placebo Study Pills (% of Pills Taken) for All Time Points, among Participants Responding to Compliance Questionnaires

eTable 5. Hazard Ratios (HR) and 95% Confidence Intervals (CI) for the Primary and Secondary Outcomes, According to Randomized Assignment to Vitamin D3 or Placebo, with Additional Censoring at Time Taking Less than 2/3 Study Pills or Taking >800 IU/d of Outside (Non-study) Vitamin D

eTable 6a. Hazard Ratios (HR) and 95% Confidence Intervals (CI) for the Primary and Secondary Outcomes, According to Randomized Assignment to Vitamin D3 or Placebo, with Additional Censoring at Incident CVD

eTable 6b. Hazard Ratios (HR) and 95% Confidence Intervals (CI) for the Primary and Secondary Outcomes, According to Randomized Assignment to Vitamin D3 or Placebo, with Additional Adjustment for CVD as a Time-Dependent Covariate

eTable 7a. Hazard Ratios (HR) and 95% Confidence Intervals (CI) for the Primary and Secondary Outcomes, According to Randomized Assignment to Vitamin D3 or Placebo, with Additional Censoring at Incident Cancer

eTable 7b. Hazard Ratios (HR) and 95% Confidence Intervals (CI) for the Primary and Secondary Outcomes, According to Randomized Assignment to Vitamin D3 or Placebo, with Additional Adjustment for Total Cancer as a Time-Dependent Covariate

Description of Results. From Sensitivity Analyses in eTable 6 and eTable 7

eTable 8. Subdistribution Hazard Models Comparing the Risk of Depression in the Vitamin D3 and Placebo Groups

eTable 9. Adjusted Differences in Change in PHQ-8 Scores Since Baseline, comparing Vitamin D3 to Placebo

eTable 10. Adjusted Means at Baseline and Mean Change (95% CI) in PHQ-8 Scores at Each Year Since Randomization Compared to Baseline, according to Vitamin D3 and Placebo Groups, Censoring PHQ-8 Scores at Date of Mood Safety Letter

Description of Results. From Sensitivity Analyses in eTable 10

eTable 11. Adjusted Means at Baseline and Mean Change (95% CI) in PHQ-8 Scores at Each Year Since Randomization Compared to Baseline, according to Vitamin D3 and Placebo Groups, Omitting Year 5 PHQ-8 Score

eTable 12. Sex-specific Total, Incident and Recurrent Rates of Depression, per 1000 Person-Years (p-y)

Description of Results. From eTable 12, Sex-Specific Depression Rates and Relation of Sex to Depression Risk

eTable 13. Total, Incident and Recurrent Rates of Depression, per 1000 Person-Years (p-y), by Vitamin D3 and Placebo Groups

eTable 14a. Association of Low Baseline 25(OH)D with Hazard Ratios and 95% Confidence Intervals for Primary Outcome (Total Depression)

eTable 14b. Association of Continuous Baseline 25(OH)D with Hazard Ratios and 95% Confidence Intervals for Primary Outcome (Total Depression)

eTable 14c. Association of Baseline 25(OH)D with Overall Mean Difference in Change (95% CI) in PHQ-8 Score Over Follow-up

Description of Results. From eTable 14

eTable 15. Means (SDs) of PHQ-8 Scores at Each Time Point, in Vitamin D3 and Placebo Groups

eFigure. Adjusted Differences in Change in Likelihood of PHQ-8 Item-Level Symptoms, comparing Vitamin D3 to Placebo

eMethods

eReferences

This supplementary material has been provided by the authors to give readers additional information about their work.

eTable 1. Detailed Baseline Characteristics of Participants, According to Randomized Vitamin D3 and Placebo Groups.*

Baseline Characteristic	Vitamin D3 Group (N=9181)	Placebo Group (N=9172)
Age, y, mean (SD)	67.5 (7.0)	67.4 (7.1)
Age groups, y, no. (%)		
50-54	319 (3.5)	333 (3.6)
55-64	2967 (32.3)	2998 (32.7)
65-74	4630 (50.4)	4601 (50.2)
≥75	1265 (13.8)	1240 (13.5)
Sex, no. (%)		
Men	4641 (50.6)	4689 (51.1)
Women	4540 (49.5)	4483 (48.9)
Race or ethnic group, no. (%) ^a	n=8999	n=8990
Non-Hispanic White	6552 (72.8)	6545 (72.8)

Baseline Characteristic	Vitamin D3 Group (N=9181)	Placebo Group (N=9172)
Black	1705 (19.0)	1702 (18.9)
Hispanic (not African American)	362 (4.0)	346 (3.9)
Asian/Pacific Islander	144 (1.6)	150 (1.7)
American Indian/Alaskan Native	78 (0.9)	72 (0.8)
Other/unknown	158 (1.8)	175(2.0)
Greater than high school education, no./total no. (%)	8078 / 9162 (88.2)	8087 / 9153 (88.4)
Income $\geq$ \$30,000 per year, no./total no. (%)	6900 / 8257 (83.6)	6885 / 8223 (83.7)
Body mass index ^b , mean (SD)	27.8 (5.4)	27.8 (5.5)
Charlson (Deyo) score ^c , no. (%)		
0 point	7744 (84.4)	7796 (85.0)
1 point	1234 (13.4)	1171 (12.8)
$\geq$ 2 points	203 (2.2)	205 (2.2)
Hypertension treated with medication, no./total no. (%)	4570 / 9129 (50.1)	4628 / 9133 (50.7)
Current use of cholesterol-lowering medication, no./total no. (%)	3366 / 9143 (36.8)	3258 / 9130 (35.7)

Baseline Characteristic	Vitamin D3 Group (N=9181)	Placebo Group (N=9172)
Diabetes, no./total no. (%)	1186 / 9164 (12.9)	1122 / 9161 (12.3)
Smoking, no. (%)	n=9134	n=9133
Never	4836 (53.0)	4902 (53.7)
Past	3727 (40.8)	3681 (40.3)
Current	571 (6.3)	550 (6.0)
Alcohol use frequency, no. (%)	n=9050	n=9042
Never/rarely	2700 (29.8)	2774 (30.7)
Monthly	663 (7.3)	656 (7.3)
Weekly	3309 (36.6)	3151 (34.9)
Daily	2378 (26.3)	2461 (27.2)
Total physical activity, MET-hours/week, median (IQR)	16.8 (5.5-32.7)	17.0 (5.6-33.5)
Current hormone use (women only), no./total no. (%) ^d	462 / 4462 (10.4)	477 / 4408 (10.8)
Current use of multivitamins, no./total no. (%)	4107 / 9046 (45.4)	4066 / 9038 (45.0)
Current use of supplemental vitamin D, no./total no. (%) ^e	4076 / 9181 (44.4)	4054 / 9172 (44.2)

Baseline Characteristic	Vitamin D3 Group (N=9181)	Placebo Group (N=9172)
Current use of supplemental calcium, no./total no. (%) ^f	1908 / 9181 (20.8)	1845 / 9172 (20.1)
Intake of foods related to vitamin D and/or n-3 fatty acids, mean (SD) ^g		
Milk, servings/day ^h (n=17927)	0.7 (0.9)	0.7 (0.9)
Other vitamin D-fortified foods, servings/day ⁱ (n=18077)	0.6 (0.8)	0.6 (0.7)
Dark-meat fish, servings/week ^j (n=18053)	1.0 (1.4)	1.0 (1.7)
Other fish and seafood, servings/week ^k (n=18064)	1.1 (1.8)	1.1 (1.7)
Baseline biomarker levels, ng/ml, mean (SD)		
25(OH)D ^l (n= 11417)	31.2 (9.8)	31.1 (10.0)
EPA (n= 11229)	0.6 (0.4)	0.6 (0.4)
DHA (n = 11237)	2.0 (0.7)	2.0 (0.7)
Location, no. (%)		
Southeast	2484 (27.1)	2547 (27.8)
Northeast	2528 (27.5)	2475 (27.0)
West	2139 (23.3)	2149 (23.4)

Baseline Characteristic	Vitamin D3 Group (N=9181)	Placebo Group (N=9172)
Midwest	2030 (22.1)	2000 (21.8)
Randomization to omega-3 Fatty Acids, no. (%)		
Active	4608 (50.2)	4563 (49.8)
Placebo	4573 (49.8)	4609 (50.3)

Abbreviations: SD, standard deviation; IQR, interquartile range; MET, Metabolic Equivalent Task; 25(OH)D, 25-hydroxyvitamin D; EPA, Eicosapentaenoic acid; DHA, Docosahexaenoic acid

* Plus - minus values are mean (SD). Percentages may not sum to 100 because of rounding. There were no significant differences between groups with regard to the baseline characteristics.

^a Race and ethnic group were reported by participants.

^b Body mass index is weight in kilograms divided by the square of height in meters. Data were missing for 2.4% of the participants.

^c The Charlson-Deyo index is a weighted comorbidity score derived from the sum of the scores for each of several major medical comorbid conditions^{1,2}. Participants were categorized as having 0, 1, or ≥ 2 points on the Charlson index.

^d Virtually all female participants are postmenopausal (>99%).

^e ≤ 800 IU/day from all supplemental sources of vitamin D combined (individual vitamin D supplements, calcium+vitamin D supplements, medications with vitamin D [e.g., Fosamax Plus D], and multivitamins)

^f ≤1200 mg/day from all supplemental sources of calcium combined

^g As assessed by a modified version of the Harvard Food Frequency Questionnaire.

^h Milk: Dairy and soy milk

ⁱ Other vitamin-D fortified foods: vitamin D-fortified cereal, vitamin D-fortified orange juice, yogurt

^j Dark-meat fish: e.g., mackerel, salmon, sardines, bluefish, swordfish; canned tuna.

^k Other fish and seafood: e.g., cod, haddock, halibut; breaded fish cakes, pieces, or fish sticks; shrimp, lobster, scallops

^l To convert 25(OH)D unit from ng/ml to nmol/l, multiply by 2.5.

eTable 2. Mean Difference in Change Since Baseline in PHQ-8 Score Comparing Vitamin D3 and Placebo Groups, According to Baseline Sub-groups.*

Group	No. of participants	Mean Difference (95% CI)	P-interaction ^a
Sex			0.14
Women	9,023	-0.03 (-0.09, 0.04)	
Men	9,330	0.04 (-0.02, 0.10)	
Age (years)			0.20
50-64	6,617	-0.01 (-0.09, 0.07)	
65-74	9,231	0.04 (-0.02, 0.10)	
≥ 75	2,505	-0.06 (-0.17, 0.05)	
Race^b			0.74
Non-Hispanic White	13097	-0.01 (-0.05, 0.04)	
Black	3407	0.04 (-0.09, 0.17)	
Other	1485	0.05 (-0.12, 0.22)	

Group	No. of participants	Mean Difference (95% CI)	P-interaction ^a
Body Mass Index^c			0.55
< 25 kg/m ²	5843	0.00 (-0.06, 0.07)	
25 - < 30 kg/m ²	7243	0.04 (-0.03, 0.10)	
≥ 30 kg/m ²	4833	-0.03 (-0.13, 0.07)	
Baseline serum 25-hydroxyvitamin D			0.96
< 20 ng/ml	1328	-0.00 (-0.17, 0.17)	
≥ 20 ng/ml	10089	0.01 (-0.05, 0.06)	
Charlson-Deyo Score			0.48
0 point	15540	0.01 (-0.03, 0.06)	
1 point	2405	0.01 (-0.13, 0.16)	
≥ 2 points	408	-0.27 (-0.65, 0.11)	
Baseline vitamin D use^d			0.26
No	10223	0.03 (-0.03, 0.09)	
Yes	8130	-0.02 (-0.08, 0.04)	

Group	No. of participants	Mean Difference (95% CI)	P-interaction ^a
Total Physical Activity (MET-hours/week)			0.96
< median of 16.9	9174	0.01 (-0.06, 0.08)	
≥ median of 16.9	9176	0.01 (-0.05, 0.06)	
Geographic Region/Latitude			0.39
Southeast	5031	-0.05 (-0.14, 0.04)	
Northeast	5003	0.03 (-0.05, 0.11)	
West	4288	0.05 (-0.03, 0.14)	
Midwest	4030	0.01 (-0.08, 0.10)	
Randomized in the n-3 fatty acids portion of trial			0.24
Active	9171	-0.02 (-0.08, 0.04)	
Placebo	9182	0.03 (-0.03, 0.09)	

Abbreviation: CI, Confidence Interval; PHQ, Patient Health Questionnaire; MET, Metabolic Equivalent Task

* Analyses were from general linear models of response profiles to estimate the means, with time modeled as indicator variables; models were controlled for age, sex, and n-3 fatty acid randomization group. Adjusted mean differences (95% CI) between the vitamin D3 and placebo groups in PHQ-8 change scores averaged across all follow-up years (years 1-5) are shown within sub-groups.

^a P-interaction is from the test of the sub-group-x-treatment-x-follow-up time interaction term in the model.

^b Race and ethnic group were reported by participants.

^c Categories of BMI were utilized for a non-prespecified/post-hoc sub-group analysis.

^d ≤ 800 IU/day from all supplemental sources of vitamin D combined (individual vitamin D supplements, calcium+vitamin D supplements, medications with vitamin D [e.g., Fosamax Plus D], and multivitamins).

Description of Results from Sub-Group Analyses in Figure 3 and eTable 2.

There were no statistically significant interactions between treatment and any sub-group indicators (main Figure 3). There was a suggestion of greater benefit for depression risk reduction among women (p-interaction=0.10), but this was not statistically significant. Estimates also were in the direction of benefit of active vitamin D3 among those with normal vs. higher BMI (p-interaction=0.06), but this was not statistically significant. Estimates were in the direction of relative risk reductions with active vitamin D3 vs. placebo among those with baseline vitamin D use (HR=0.87, 95% CI: 0.73-1.04) and sufficient 25(OH)D blood level of ≥ 20 ng/ml (HR=0.89, 95% CI: 0.77-1.04), but these were not statistically significant. There was no indication of varying effects of active vitamin D3 on depression risk by age, race, geographic location or other factors of interest.

eTable 2 shows results for the effects of vitamin D3 vs. placebo on the overall mean differences in mood score change since baseline, by sub-groups. There was no variation by sub-groups in the effect of vitamin D3 on overall mean differences in mood score change since baseline. Tests of statistical interactions in sub-group analyses were not adjusted for multiple comparisons.

eTable 3. Adjusted Means at Baseline and Mean Change (95% CI) in PHQ-8 Scores at Each Year Since Randomization Compared to Baseline, according to Vitamin D3 and Placebo Groups, Censoring PHQ-8 Scores after Initiation of Antidepressants.^a

PHQ-8 Score	Vitamin D3 Group		Placebo Group		Mean Difference (95% CI) ^b	P-value	P-interaction
	Number of Participants	Adjusted mean (95% CI)	Number of Participants	Adjusted mean (95% CI)			
Baseline	9181	1.08 (1.05, 1.11)	9172	1.13 (1.09, 1.16)	--	--	0.88
Year 1 vs Baseline	8479	0.02 (-0.02, 0.05)	8438	0.03 (-0.01, 0.06)	-0.01 (-0.06, 0.04)	0.71	
Year 2 vs Baseline	8282	0.06 (0.02, 0.10)	8260	0.04 (-0.00, 0.08)	0.02 (-0.03, 0.08)	0.45	
Year 3 vs Baseline	8032	0.07 (0.03, 0.11)	7993	0.07 (0.02, 0.11)	0.01 (-0.05, 0.07)	0.82	
Year 4 vs Baseline	7590	0.04 (0.00, 0.08)	7455	0.05 (0.01, 0.09)	-0.01 (-0.07, 0.05)	0.80	
Year 5 vs Baseline	5181	0.17 (0.12, 0.22)	5096	0.15 (0.10, 0.20)	0.02 (-0.05, 0.09)	0.64	
Average (across Years 1-5) vs Baseline	9181		9172		0.00 (-0.04, 0.05)	0.89	

Abbreviation: CI, Confidence Interval; PHQ, Patient Health Questionnaire

^a Analyses were from general linear models of response profiles to estimate the means, with time modeled as indicator variables; models were controlled for age, sex, and fish oil randomization group. Adjusted means (95% CI) within each treatment group are

shown at baseline and adjusted mean differences in change (95% CI) within each treatment group are shown for each follow-up time point. P-interaction is from the 5-degree-freedom test of the treatment-x-time interaction term in the model.

^b Mean differences in change comparing vitamin D3 and placebo groups; the last row shows the adjusted mean difference (95% CI) between the vitamin D3 and placebo groups in PHQ-8 change scores averaged across all follow-up years (years 1-5 vs. baseline).

eTable 4. Participant-Reported Adherence with the Vitamin D3 and Placebo Study Pills (% of Pills Taken) for All Time Points, among Participants Responding to Compliance Questionnaires.

	Vitamin D3	Placebo
Baseline	9181 / 9181 (100.0)	9172 / 9172 (100.0)
Year 1	8237 / 8688 (94.8)	8128 / 8590 (94.6)
Year 2	7793 / 8448 (92.2)	7681 / 8341 (92.1)
Year 3	7485 / 8180 (91.5)	7367 / 8097 (91.0)
Year 4	7103 / 7771 (91.4)	6920 / 7644 (90.5)
Year 5	4665 / 5148 (90.6)	4524 / 5036 (89.8)

eTable 5. Hazard Ratios (HR) and 95% Confidence Intervals (CI) for the Primary and Secondary Outcomes, According to Randomized Assignment to Vitamin D3 or Placebo, with Additional Censoring at Time Taking Less than 2/3 Study Pills or Taking >800 IU/d of Outside (Non-study) Vitamin D.*

Outcome	Vitamin D3 Group (n=9181)	Placebo Group (n=9172)	HR (95% CI)	P-value
	<i>no. of participants with event</i>			
Total Depression ^a	468	486	0.94 (0.83-1.07)	0.35
Incident Depression ^b	350	362	0.94 (0.81-1.09)	0.40
Recurrent Depression ^b	118	124	0.99 (0.77-1.28)	0.97

* Analyses were from Cox regression models that were controlled for age, sex, and marine-3 fatty acid randomization group. Analyses were not adjusted for multiple comparisons.

^a Primary Outcome: Depression is a composite endpoint comprising reported presence of clinician diagnosis of depression, treatment for depression and/or symptoms above the validated cutoff for major depression on the PHQ-8 ($\text{PHQ-8} \geq 10$); total depression consists of all incident and recurrent depression combined.

^b Secondary Outcome: Incident and recurrent depression were secondary outcomes. Incident depression was defined as depression cases that occurred among those with no past history of depression; recurrent depression was defined as depression cases that occurred among those with past history of depression, but not under treatment or active in the past 2 years.

eTable 6a. Hazard Ratios (HR) and 95% Confidence Intervals (CI) for the Primary and Secondary Outcomes, According to Randomized Assignment to Vitamin D3 or Placebo, with Additional Censoring at Incident CVD.*

Outcome	Vitamin D3 Group (n=9181)	Placebo Group (n=9172)	HR (95% CI)	P-value
	no. of participants with event			
Total Depression ^a	598	606	0.99 (0.88-1.10)	0.80
Incident Depression ^b	453	446	1.01 (0.89-1.15)	0.87
Recurrent Depression ^b	145	160	0.95 (0.76-1.19)	0.64

* Analyses were from Cox regression models that were controlled for age, sex, and marine-3 fatty acid randomization group. Analyses were not adjusted for multiple comparisons.

^a Primary Outcome: Depression is a composite endpoint comprising reported presence of clinician diagnosis of depression, treatment for depression and/or symptoms above the validated cutoff for major depression on the PHQ-8 (PHQ-8 $\geq$ 10); total depression consists of all incident and recurrent depression combined.

^b Secondary Outcome: Incident and recurrent depression were secondary outcomes. Incident depression was defined as depression cases that occurred among those with no past history of depression; recurrent depression was defined as depression cases that occurred among those with past history of depression, but not under treatment or active in the past 2 years.

eTable 6b. Hazard Ratios (HR) and 95% Confidence Intervals (CI) for the Primary and Secondary Outcomes, According to Randomized Assignment to Vitamin D3 or Placebo, with Additional Adjustment for CVD as a Time-Dependent Covariate.*

Outcome	Vitamin D3 Group	Placebo Group	HR (95% CI)	P-value
	(n = 9181)	(n = 9172)		
	<i>no. of participants with event</i>			
Total Depression ^a	609	625	0.97 (0.87-1.09)	0.60
Incident Depression ^b	459	461	0.99 (0.87-1.13)	0.87

Outcome	Vitamin D3 Group	Placebo Group	HR (95% CI)	P-value
	(n = 9181)	(n = 9172)		
	no. of participants with event			
Recurrent Depression ^b	150	164	0.95 (0.76-1.18)	0.63

* Analyses were from Cox regression models that were controlled for age, sex, time-dependent CVD variable and marine-3 fatty acid randomization group. Analyses were not adjusted for multiple comparisons.

^a Primary Outcome: Depression is a composite endpoint comprising reported presence of clinician diagnosis of depression, treatment for depression and/or symptoms above the validated cutoff for major depression on the PHQ-8 (PHQ-8 $\geq$ 10); total depression consists of all incident and recurrent depression combined.

^b Secondary Outcome: Incident and recurrent depression were secondary outcomes. Incident depression was defined as depression cases that occurred among those with no past history of depression; recurrent depression was defined as depression cases that occurred among those with past history of depression, but not under treatment or active in the past 2 years.

eTable 7a. Hazard Ratios (HR) and 95% Confidence Intervals (CI) for the Primary and Secondary Outcomes, According to Randomized Assignment to Vitamin D3 or Placebo, with Additional Censoring at Incident Cancer.*

Outcome	Vitamin D3 Group	Placebo Group	HR (95% CI)	P-value
	(n=9181)	(n=9172)		
	no. of participants with event			
Total Depression ^a	590	602	0.98 (0.87-1.10)	0.70
Incident Depression ^b	445	441	1.00 (0.88-1.14)	0.96
Recurrent Depression ^b	145	161	0.94 (0.75-1.18)	0.61

* Analyses were from Cox regression models that were controlled for age, sex, and marine-3 fatty acid randomization group. Analyses were not adjusted for multiple comparisons.

^a Primary Outcome: Depression is a composite endpoint comprising reported presence of clinician diagnosis of depression, treatment for depression and/or symptoms above the validated cutoff for major depression on the PHQ-8 (PHQ-8 $\geq$ 10); total depression consists of all incident and recurrent depression combined.

^b Secondary Outcome: Incident and recurrent depression were secondary outcomes. Incident depression was defined as depression cases that occurred among those with no past history of depression; recurrent depression was defined as depression cases that occurred among those with past history of depression, but not under treatment or active in the past 2 years.

eTable 7b. Hazard Ratios (HR) and 95% Confidence Intervals (CI) for the Primary and Secondary Outcomes, According to Randomized Assignment to Vitamin D3 or Placebo, with Additional Adjustment for Total Cancer as a Time-Dependent Covariate.*

Outcome	Vitamin D3 Group	Placebo Group	HR (95% CI)	P-value
	(n = 9181)	(n = 9172)		
	no. of participants with event			
Total Depression ^a	609	625	0.97 (0.87-1.09)	0.62
Incident Depression ^b	459	461	0.99 (0.87-1.13)	0.88
Recurrent Depression ^b	150	164	0.95 (0.76-1.19)	0.67

* Analyses were from Cox regression models that were controlled for age, sex, time-dependent malignant cancer and marine-3 fatty acid randomization group. Analyses were not adjusted for multiple comparisons.

^a Primary Outcome: Depression is a composite endpoint comprising reported presence of clinician diagnosis of depression, treatment for depression and/or symptoms above the validated cutoff for major depression on the PHQ-8 ($\text{PHQ-8} \geq 10$); total depression consists of all incident and recurrent depression combined.

^b Secondary Outcome: Incident and recurrent depression were secondary outcomes. Incident depression was defined as depression cases that occurred among those with no past history of depression; recurrent depression was defined as depression cases that occurred among those with past history of depression, but not under treatment or active in the past 2 years.

Description of Results from Sensitivity Analyses in eTable 6 and eTable 7.

There were no differences in results with additional censoring at developing of parent trial CVD endpoints (total depression HR, 95% CI: 0.99, 0.88-1.10). While development of CVD was strongly associated with 3-fold risk of total depression (HR, 95% CI: 2.96, 2.05-4.27), incident depression (HR, 95% CI: 2.72, 1.76-4.21) and recurrent depression (HR, 95% CI: 2.85, 1.46-5.55), there were no differences in results for the effect of vitamin D3 on depression risk when including CVD as a time-updated covariate (total depression HR, 95% CI: 0.97, 0.87-1.09). Similarly, there were no differences in results when censoring or adjusting for time-updated parent trial cancer endpoints. Development of cancer was not statistically significantly associated with total (HR, 95% CI: 1.21, 0.89-1.65), incident (HR, 95% CI: 1.28, 0.91-1.81) or recurrent depression (HR, 95% CI: 0.97, 0.48-1.97).

eTable 8. Subdistribution Hazard Models Comparing the Risk of Depression in the Vitamin D3 and Placebo Groups.*

Outcome	No. of participants	HR (95% CI)	P-value
Total Depression ^a	18,353	0.97 (0.87-1.09)	0.60
Incident Depression ^b	16,657	0.99 (0.87-1.13)	0.87
Recurrent Depression ^b	1696	0.95 (0.76-1.18)	0.62

Abbreviation: HR, Hazard Ratio; CI, confidence interval

*The adjusted HRs were computed from the Fine and Gray subdistribution hazard models. Analyses used the Fine-Gray competing risks approach; death from any cause was treated as a competing rather than censored event. Results from the subdistribution hazard models are shown for total, incident and recurrent depression or clinically relevant depressive symptoms.

^a Primary Outcome: Depression is a composite endpoint comprising reported presence of clinician diagnosis of depression, treatment for depression and/or symptoms above the validated cutoff for major depression on the PHQ-8 (PHQ-8 $\geq$ 10); total depression consists of all incident and recurrent depression combined.

^b Secondary Outcome: Incident and recurrent depression were secondary outcomes. Incident depression was defined as depression cases that occurred among those with no past history of depression; recurrent depression was defined as depression cases that occurred among those with past history of depression, but not under treatment or active in the past 2 years.

eTable 9. Adjusted Differences in Change in PHQ-8 Scores Since Baseline, comparing Vitamin D3 to Placebo.^a

Effect	Vitamin D3 Group, N	Placebo Group, N	Rate Ratio (95% CI)	P-Value
Year 1 vs Baseline	8534	8486	1.00 (0.95, 1.05)	0.92
Year 2 vs Baseline	8381	8344	1.03 (0.98, 1.08)	0.22
Year 3 vs Baseline	8176	8112	1.02 (0.96, 1.07)	0.57
Year 4 vs Baseline	7763	7603	1.00 (0.95, 1.05)	0.87
Year 5 vs Baseline	5316	5231	1.03 (0.97, 1.09)	0.30
Average over Years 1-5 vs Baseline	9181	9172	1.01 (0.97, 1.05)	0.51

Abbreviation: CI, Confidence Interval; PHQ, Patient Health Questionnaire

^a Analyses were from repeated measures negative binomial regression models, with follow-up time modeled as an indicator; models were controlled for age, sex, and n-3 fatty acid randomization group. Results show rate ratios (RRs) and 95% confidence intervals (95% CIs), which reflect percent differences in the change in severity on the PHQ-8 score comparing vitamin D3 to placebo treatment group. RRs are shown for each follow-up time point, and for the average over all follow-up. Results show no significant differences between the treatment groups in change in PHQ-8 scores since baseline.

eTable 10. Adjusted Means at Baseline and Mean Change (95% CI) in PHQ-8 Scores at Each Year Since Randomization Compared to Baseline, according to Vitamin D3 and Placebo Groups, Censoring PHQ-8 Scores at Date of Mood Safety Letter.^a

PHQ-8 Score	Vitamin D3 Group		Placebo Group		Mean Difference (95% CI) ^b	P-value	P- interaction
	Number of Participants	Adjusted mean (95% CI)	Number of Participants	Adjusted mean (95% CI)			
Baseline	9181	1.08 (1.05, 1.11)	9172	1.13 (1.09, 1.16)	--	--	0.63
Year 1 vs Baseline	8534	0.03 (-0.01, 0.07)	8486	0.03 (-0.00, 0.07)	-0.00 (-0.06, 0.05)	0.86	
Year 2 vs Baseline	8349	0.07 (0.04, 0.11)	8311	0.05 (0.01, 0.09)	0.02 (-0.03, 0.08)	0.42	
Year 3 vs Baseline	8115	0.11 (0.07, 0.15)	8054	0.08 (0.04, 0.12)	0.02 (-0.04, 0.08)	0.43	
Year 4 vs Baseline	7671	0.08 (0.04, 0.13)	7517	0.10 (0.05, 0.14)	-0.01 (-0.07, 0.05)	0.69	
Year 5 vs Baseline	5239	0.24 (0.19, 0.29)	5159	0.21 (0.16, 0.26)	0.03 (-0.04, 0.10)	0.39	
Average (across Years 1-5) vs Baseline	9181		9172		0.01 (-0.04, 0.05)	0.71	

Abbreviation: PHQ, patient health questionnaire; CI, confidence interval

^a Analyses were from general linear models of response profiles to estimate the means, with time modeled as indicator variables; models were controlled for age, sex, and n-3 fatty acid randomization group. Adjusted means (95% CI) within each treatment group are shown at baseline and adjusted mean differences in change (95% CI) within each treatment group are shown for each follow-up time point. P-interaction is from the 5-degree-freedom test of the treatment-x-time interaction term in the model.

^b Mean differences in change comparing vitamin D3 and placebo groups; the last row shows the adjusted mean difference (95% CI) between the vitamin D3 and placebo groups in PHQ-8 change scores averaged across all follow-up years (years 1-5 vs. baseline).

Description of Results from Sensitivity Analyses in eTable 10.

As described in eMethods, enhanced follow-up procedures were instituted for those with elevated PHQ-8 scores. Participants with elevated PHQ-8 scores (≥ 10 algorithm cutoff) at baseline and follow-up were contacted via mailed letters, where there was no current self-report by the participant of both recent diagnosis and treatment of depression. Among all participants who scored PHQ-8 ≥ 15 , the same letters were sent regardless of recent self-reported diagnosis or treatment for depression. It was recognized that receipt of a letter intended to raise a participant's awareness of mood problems may influence his or her self-report of mood on a subsequent questionnaire, and this may have potential to bias results. Thus, in this analysis PHQ-8 scores were censored after the date that a mood safety letter was sent (i.e., PHQ-8 responses that occurred after the send date of a mood safety letter did not contribute to the outcome). Results showed that estimates observed in this sensitivity analysis are similar to those observed in the primary analysis.

eTable 11. Adjusted Means at Baseline and Mean Change (95% CI) in PHQ-8 Scores at Each Year Since Randomization Compared to Baseline, according to Vitamin D3 and Placebo Groups, Omitting Year 5 PHQ-8 Score.^a

PHQ-8 Score	Vitamin D3 Group		Placebo Group		Mean Difference (95% CI) ^b	P-value	P- interaction
	Number of Participants	Adjusted mean (95% CI)	Number of Participants	Adjusted mean (95% CI)			
Baseline	9181	1.08 (1.05, 1.11)	9172	1.13 (1.09, 1.16)	--	--	0.73
Year 1 vs Baseline	8534	0.03 (-0.01, 0.07)	8486	0.03 (-0.01, 0.07)	-0.01 (-0.06, 0.05)	0.84	
Year 2 vs Baseline	8381	0.07 (0.03, 0.11)	8344	0.04 (0.00, 0.08)	0.03 (-0.03, 0.08)	0.36	
Year 3 vs Baseline	8176	0.09 (0.05, 0.13)	8112	0.07 (0.03, 0.11)	0.01 (-0.05, 0.07)	0.65	
Year 4 vs Baseline	7763	0.06 (0.02, 0.10)	7603	0.07 (0.03, 0.11)	-0.01 (-0.07, 0.05)	0.72	
Average (across Years 1-4) vs Baseline	9181		9172		0.00 (-0.04, 0.05)	0.83	

Abbreviation: CI, Confidence Interval; PHQ, Patient Health Questionnaire

^a Analyses were from general linear models of response profiles to estimate the means, with time modeled as indicator variables; models were controlled for age, sex, and n-3 fatty acid randomization group. Adjusted means (95% CI) within each treatment group

are shown at baseline and adjusted mean differences in change (95% CI) within each treatment group are shown for each follow-up time point. P-interaction is from the test of the treatment-x-time interaction term in the model.

^b Mean differences in change comparing vitamin D3 and placebo groups; the last row shows the adjusted mean difference (95% CI) between the vitamin D3 and placebo groups in PHQ-8 change scores averaged across all follow-up years (years 1-4 vs. baseline).

eTable 12. Sex-specific Total, Incident and Recurrent Rates of Depression, per 1000 Person-Years (p-y).

Men			
Depression Outcome	Number of Participants	Cases	Per 1000 p-y
Total depression ^a	9330	548	11.36
Incident depression ^b	8642	426	9.49
Recurrent depression ^b	688	122	36.27
Women			
Depression Outcome	Number of Participants	Cases	Per 1000 p-y
Total depression ^a	9023	686	14.97
Incident depression ^b	8015	494	12.04
Recurrent depression ^b	1008	192	40.01

^a Primary Outcome: Depression is a composite endpoint comprising reported presence of clinician diagnosis of depression, treatment for depression and/or symptoms above the validated cutoff for major depression on the PHQ-8 (PHQ-8 $\geq$ 10); total depression consists of all incident and recurrent depression combined.

^b Secondary Outcome: Incident and recurrent depression were secondary outcomes. Incident depression was defined as depression cases that occurred among those with no past history of depression; recurrent depression was defined as depression cases that occurred among those with past history of depression, but not under treatment or active in the past 2 years.

Description of Results from eTable 12, Sex-Specific Depression Rates and Relation of Sex to Depression Risk.

We examined total and sex-specific case rates for incident and recurrent depression. The total depression rate was 11.4/1,000 p-y in males and 15.0/1,000 p-y in females. The incidence rate was 10.7/1,000 p-y (9.5 and 12.0 per 1,000 p-y in males and females, respectively), and the recurrence rate was 38.5/1,000 p-y (36.3 and 40.0 per 1,000 p-y in males and females, respectively). Thus, both rates and sex ratios of late-life depression incidence and recurrence were comparable to those reported elsewhere³. As sex/gender is one of the most robust known predictors of depression risk across age groups, we provide the estimate for the association of women vs. men with risk of total depression, from the primary results model: hazard ratio=1.34, 95% CI: 1.19, 1.50.

eTable 13. Total, Incident and Recurrent Rates of Depression, per 1000 Person-Years (p-y), by Vitamin D3 and Placebo Groups.

Vitamin D3 group			
Depression Outcome	Number of Participants	Cases	Per 1000 p-y
Total depression ^a	9181	609	12.95
Incident depression ^b	8350	459	10.66
Recurrent depression ^b	831	150	37.58
Placebo group			
Depression Outcome	Number of Participants	Cases	Per 1000 p-y
Total depression ^a	9172	625	13.29
Incident depression ^b	8307	461	10.76
Recurrent depression ^b	865	164	39.32

^a Primary Outcome: Depression is a composite endpoint comprising reported presence of clinician diagnosis of depression, treatment for depression and/or symptoms above the validated cutoff for major depression on the PHQ-8 (PHQ-8 $\geq$ 10); total depression consists of all incident and recurrent depression combined.

^b Secondary Outcome: Incident and recurrent depression were secondary outcomes. Incident depression was defined as depression cases that occurred among those with no past history of depression; recurrent depression was defined as depression cases that occurred among those with past history of depression, but not under treatment or active in the past 2 years.

eTable 14a. Association of Low Baseline 25(OH)D with Hazard Ratios and 95% Confidence Intervals for Primary Outcome (Total Depression).*

Exposure	N	HR (95% CI)	P-value
Sufficient Vitamin D	10089	1.00 (Ref)	N/A
Low Vitamin D	1328	1.08 (0.87-1.35)	0.48

eTable 14b. Association of Continuous Baseline 25(OH)D with Hazard Ratios and 95% Confidence Intervals for Primary Outcome (Total Depression).*

Exposure	N	HR (95% CI)	P-value
Per 10 ng/ml increase in 25(OH)D	11417	1.00 (0.93-1.08)	0.94

* Sample n=11,417 with available 25(OH)D levels in this study. Analyses were from Cox regression models that were controlled for age, sex, and vitamin D3 and n-3 fatty acid randomization groups. Low vitamin D is considered 25(OH)D <20 ng/ml.

eTable 14c. Association of Baseline 25(OH)D with Overall Mean Difference in Change (95% CI) in PHQ-8 Score Over Follow-up.*

Exposure	Mean Difference (95% CI)	P-value
Low Vitamin D	-0.04 (-0.13, 0.06)	0.45
Per 10 ng/ml increase in 25(OH)D	0.02 (-0.01, 0.05)	0.14

Abbreviation: HR, Hazard Ratio; CI, Confidence Interval; PHQ, Patient Health Questionnaire

* Sample n=11,417 with available 25(OH)D levels in this study. Analyses were from general linear models of response profiles, controlling for age, sex, and vitamin D3 and n-3 fatty acid randomization groups. Low vitamin D is considered 25(OH)D <20 ng/ml.

Description of Results from eTable 14.

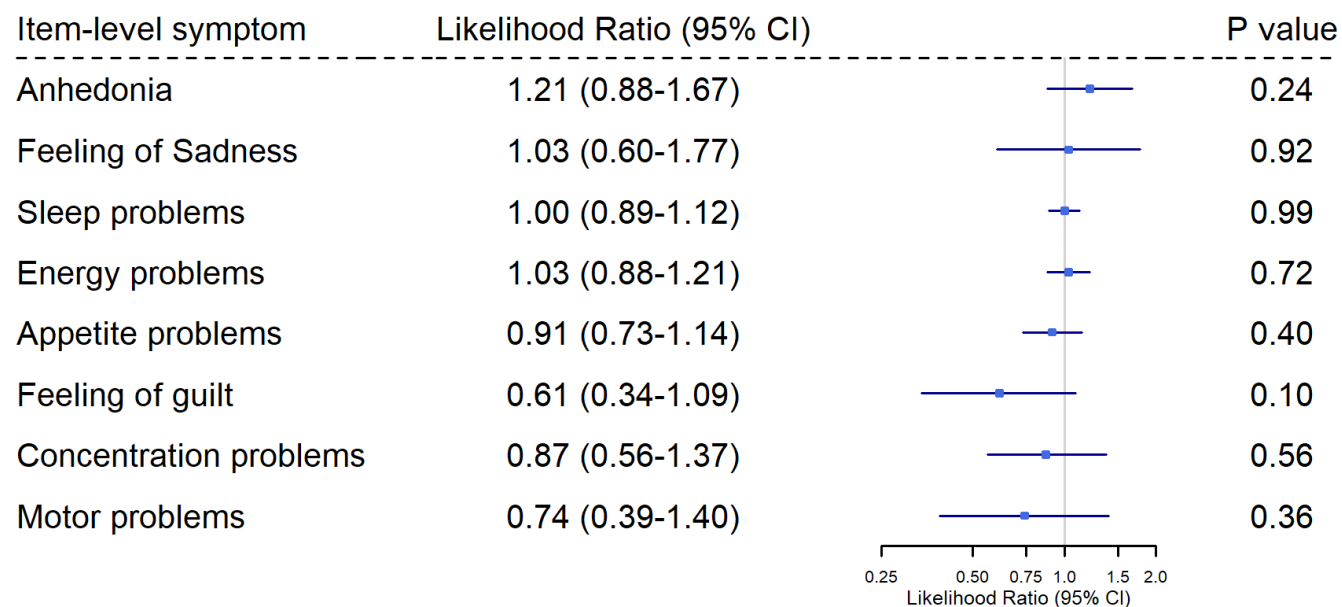
There was no association between 25(OH)D level and total depression risk (adjusted HR, 95% CI: 1.00, 0.93-1.08) or between low vitamin D (25(OH)D <20 ng/ml) and depression risk (adjusted HR, 95% CI: 1.08, 0.87-1.35), although estimates were slightly in the direction of higher depression risk with low/insufficient 25(OH)D. In models of the associations of 25(OH)D levels with change in mood scores, there were significantly lower depressive symptoms scores associated with each 10 ng/ml increment in baseline 25(OH)D level at all follow-up time points; however, there was no significant difference in the overall change since baseline (eTable 14c). Similarly, participants with low (<20 ng/ml) vs. sufficient baseline 25(OH)D level had significantly worse depressive symptoms scores at the follow-up time points; however, there was no significant difference in the overall change since baseline comparing the low vs. sufficient 25(OH)D groups (eTable 14c). Thus, while baseline 25(OH)D status was significantly associated with initial level of mood scores, there were no significant associations between 25(OH)D and change in mood scores.

eTable 15. Means (SDs) of PHQ-8 Scores at Each Time Point, in Vitamin D3 and Placebo Groups.

Time	Vitamin D3	Placebo
Baseline	1.08 (1.60)	1.12 (1.63)
Year 1	1.10 (1.89)	1.14 (1.95)
Year 2	1.12 (1.98)	1.14 (1.97)
Year 3	1.12 (1.97)	1.16 (2.02)
Year 4	1.09 (1.91)	1.14 (2.01)
Year 5	1.23 (2.06)	1.23 (2.10)

Abbreviation: PHQ, Patient Health Questionnaire

eFigure. Adjusted Differences in Change in Likelihood of PHQ-8 Item-Level Symptoms, comparing Vitamin D3 to Placebo.^a



Abbreviation: CI, Confidence Interval; PHQ, Patient Health Questionnaire

^a Analyses were from repeated measures logistic regression models, with follow-up time modeled as an indicator; models were controlled for age, sex, and n-3 fatty acid randomization group. Results show likelihood ratios and 95% confidence intervals (95% CIs), which reflect differences in the change in likelihood of burden from each PHQ-8 item-level symptom, comparing vitamin D3 to placebo treatment group. Differences reflect the average effect over all follow-up times since baseline.

eMethods

Changes to the study protocol

Changes occurred in the methods and statistical analysis plan (SAP) and protocol (see Supplement 1) since this study was funded in 2010.

(1) The final number of participants in the Vitamin D and Omega-3 Trial (VITAL) was 25,871 – not the 20,000 initially planned. When VITAL was designed in 2008, the planned enrollment was $n=20,000$, with a goal of 25% (or $n=5,000$) participation by Black/African-American adults. The target sample size was later increased to 26,000 participants to ensure robust minority participation, including attainment of the goal of at least $n=5,000$ Black participants.

(2) The eligible age ranges in the parent VITAL trial were modified. Lower age limits of 60 years in men and 65 years in women had originally been selected because these were observed to represent the ages at which chronic disease rates begin to increase substantially. The lower age limits were later modified (50 years and 55 years among men and women, respectively) to better reflect health disparities of earlier morbidity and mortality from CVD among Black and Hispanic adults, especially men.

(3) Because the grant that supported this study entered a 5-year renewal period while the randomized trial portion was still underway, there were additional planned analyses related to renewal period aims listed in the published protocol; however, these analyses were not related to the main trial agent-related aims from the 2010 grant submission.

(4) It was originally planned that CMS (Centers for Medicare & Medicaid Services) data would play a larger role in supplementing depression follow-up information. However, such linkage

was available in ~70% of participants; thus, this was not pursued to avoid introducing bias in ascertainment of outcomes.

(5) An earlier version of the SAP and ClinicalTrials.gov record (in 2012) listed “no history of dysthymia with active dysthymic symptoms in the past one year” as an eligibility criterion; this was erroneous and was revised at the next ClinicalTrials.gov record update in 2013, and the study criteria also appeared correctly in the peer-reviewed published trial protocol.

(6) It was possible to include more data in some areas than had been initially projected/planned at the time of study funding in 2010. First, the Patient Health Questionnaire-8 (PHQ-8) was featured on each of the annual questionnaires – for a total of 6 repeated measures (baseline/year 0 and years 1-5) – while the original SAP included the PHQ-8 only in years 0, 1, 3 and 5.

Second, through resources in the main VITAL trial, blood levels of 25(OH)D and fatty acids were measured in all participants with blood samples collected at baseline, not just a subset, and the moderator analyses accordingly were able to involve a larger sample.

(7) As described in the main text of the manuscript, this paper included post-hoc analyses that were not part of the original SAP.

(8) Not all outcomes ascertained in this study were reported in this paper. Effects of n-3 fatty acids on depression prevention were not presented here, as the extensive scope of this work would not permit inclusion in appropriate detail in this paper; separate publication is planned. Results were also not presented for the secondary aims addressing vitamin D3 for indicated and selective prevention of depression among the subset of ~1,000 Boston-based participants; separate publication of outcomes in the local Clinical and Translational Science Center (CTSC) sub-cohort (see below) is planned.

Enhanced Safety Follow-up Procedures

As detailed in the Institutional Review Board (IRB) safety protocol and published study protocol, we recognized the need for enhanced safety measures in the unique context of a large-scale study in the general population. Our approach balanced important considerations. Many participants may not recognize symptoms of depression as warranting further evaluation – in the way they might recognize the importance of symptoms such as chest pain, shortness of breath or bleeding, for example. On the other hand, some participants with elevated mood scores on the PHQ-8 may not actually have depression and could become unduly alarmed or feel stigmatized by being treated differently due to these symptoms compared to any number of other potential health concerns. Furthermore, given that several weeks of processing time are routinely required to compute score results for each participant in a study of this scope, it was necessary to avoid creating an expectation of immediate follow-up of mood items noted by participants on the questionnaires. Following careful consultation with local Human Subjects Research committee leaders and an independent research ethics team, we developed in this study a unique procedure to ensure participants are provided timely information and education about mood and identifying the means for evaluation and treatment, while minimizing potential issues of stigmatization or exceptionalization.

Thus, we instituted enhanced follow-up procedures for participants with elevated PHQ-8 scores^{4,5} in order to provide further assurance of safety. We contacted participants with elevated PHQ-8 scores (≥ 10 algorithm cutoff for major depression) at baseline and follow-up by sending mailed letters, whenever there was no current self-report by the participant of both recent diagnosis and treatment of depression. Among all participants who scored PHQ-8 ≥ 15 (validated cutoff for moderately severe or severe symptoms), we sent these same letters, regardless of

recent self-reported diagnosis or treatment for depression. The objective was to ensure that participants were educated about the importance of discussing their mood with their local health providers and, thus, initiate a means for ongoing evaluation and/or treatment as necessary; thus, among participants who were randomized and continued in the study, we only sent these letters when participants first reported above-threshold scores. As part of the informed consent procedure within the parent trial, participants had already been notified that study investigators might be contacting them directly with regard to following up on development of endpoints, to clarify their questionnaire responses and also to obtain releases to contact their primary care providers. Study staff, including the research assistants and programmers, worked with parent trial team members to send out the letters and to perform data tracking of the letters, as well as to field and respond to comments, questions or responses from participants in response to the letters. All replies, comments and correspondence from participants in response to the mood safety letters were tracked and recorded in detail, and the number of letters sent and participant feedback responses were entered into the IRB record at each annual continuing review for IRB review and approval throughout the duration of the study.

Concordance analyses in the CTSC sub-cohort

As detailed elsewhere⁶, this study featured in-person assessments using gold-standard psychiatric diagnostic interviews among the 1,054 participants who took part in the CTSC component of the parent trial. Study participants were administered the Mini-International Neuropsychiatric Interview (MINI) for the DSM (Diagnostic and Statistical Manual)-IV⁷, which is validated for time-efficient⁸ determinations of safety and diagnostic status for both eligibility and outcome purposes. The DSM-IV was utilized in this study, as the protocol was submitted

and approved by the funding agency in 2010 – three years prior to the publication of the DSM-5. Within the CTSC, administration of the MINI, along with the nine-item version of the PHQ, allowed for real-time assessments of symptoms that may indicate safety risks (i.e., suicidality, manic symptoms, or psychotic symptoms). Study psychiatrists evaluated persons immediately if they were presenting with such symptoms and determined their safety and ability to participate further in this study. Furthermore, PHQ-9 scores obtained as part of patient-reported outcomes measurement in the CTSC were compared directly to the PHQ-8 scores from the mailed questionnaires. The vast majority of participants completed their baseline parent trial questionnaires and CTSC assessments for this study within one month, and most within 2 weeks. The intraclass correlation for PHQ-8 scores was 0.63 (95% CI=0.59-0.67) among 1,053 participants who completed both in-person, locally administered PHQ-9 forms and nationally administered PHQ-8 questionnaires. In addition, there was 86% agreement with incidence and recurrence eligibility in this study with classification of having no prior or active major depression or dysthymia history among the CTSC sub-cohort in this study with gold-standard interviews using the MINI.

eReferences

1. Charlson ME, Pompei P, Ales KL, MacKenzie CR. A new method of classifying prognostic comorbidity in longitudinal studies: development and validation. *J Chronic Dis*. 1987;40(5):373-383.
2. Deyo RA, Cherkin DC, Ciol MA. Adapting a clinical comorbidity index for use with ICD-9-CM administrative databases. *J Clin Epidemiol*. 1992;45(6):613-619.
3. Luijendijk HJ, van den Berg JF, Dekker MJ, et al. Incidence and recurrence of late-life depression. *Arch Gen Psychiatry*. 2008;65(12):1394-1401.
4. Kroenke K, Spitzer RL, Williams JB. The PHQ-9: validity of a brief depression severity measure. *J Gen Intern Med*. 2001;16(9):606-613.
5. Kroenke K, Strine TW, Spitzer RL, Williams JB, Berry JT, Mokdad AH. The PHQ-8 as a measure of current depression in the general population. *J Affect Disord*. 2009;114(1-3):163-173.
6. Okereke OI, Reynolds CF, Mischoulon D, et al. The VITamin D and OmegA-3 TriaL-Depression Endpoint Prevention (VITAL-DEP): Rationale and design of a large-scale ancillary study evaluating vitamin D and marine omega-3 fatty acid supplements for prevention of late-life depression. *Contemp Clin Trials*. 2018;68:133-145.
7. Sheehan DV, Lecrubier Y, Sheehan KH, et al. The Mini-International Neuropsychiatric Interview (M.I.N.I.): the development and validation of a structured diagnostic psychiatric interview for DSM-IV and ICD-10. *J Clin Psychiatry*. 1998;59 Suppl 20:22-33;quiz 34-57.
8. van't Veer-Tazelaar PJ, van Marwijk HW, van Oppen P, et al. Stepped-care prevention of anxiety and depression in late life: a randomized controlled trial. *Arch Gen Psychiatry*. 2009;66(3):297-304.